AI-Assisted Tumour Board Emulation for Multimodal Clinical Decision Support

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Student's Declaration

I hereby declare that the work presented in the report entitled **"AI-Assisted Tumour Board Emulation for Multimodal Clinical Decision Support"** submitted by me for the complete fulfillment of the requirements for the degree of *Bachelor of Technology* in *Computer Science & Social Sciences* at Indraprastha Institute of Information Technology, Delhi, is an authentic record of my work carried out under guidance of **Dr. Jainendra Shukla**. Due acknowledgements have been given in the report to all material used. This work has not been submitted anywhere else for the reward of any other degree.

Aayush Mishra
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Place & Date: IIIT Delhi, 21^{th} **April, 2026**

Certificate

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

....
Dr. Jainendra Shukla

Place & Date: Delhi, ...

Abstract

Tumour boards are multidisciplinary clinical meetings in which oncologists, radiologists, pathologists, surgeons, and hepatology specialists collaboratively review patient cases to determine optimal treatment pathways. While tumour boards are essential for improving cancer outcomes, they remain resource-intensive, time-dependent, and often inaccessible in resource-constrained settings, leading to delays, fragmented reasoning, and variability in clinical decisions. As cancer care becomes increasingly multimodal, involving laboratory timelines, CT/MRI imaging, pathology reports, and longitudinal treatment records, the manual aggregation and interpretation of such data impose significant cognitive and operational burdens on clinicians.

To address these challenges, this project presents Samavet AI, an AI-assisted tumour board emulation system designed to synthesise multimodal patient data and generate structured, explainable, tumour-board-style clinical summaries and treatment recommendations under physician supervision. The system comprises a secure, web-based dashboard and patient entity management platform built using React and Python-based backend services, supporting structured data ingestion, PDF-based record parsing, and schema-driven patient representation aligned with clinical standards.

The architecture adopts a modular multi-agent framework, integrating modality-specific agents (clinical, radiology, pathology), a strategy planning agent, specialist reasoning agents, and an orchestration layer that produces unified tumour board reports. In the extended phase of this work, the system was enhanced with a knowledge graph–based reasoning engine leveraging ReAct-style planning and rule-based decision routing, enabling context-aware agent invocation and improved interpretability. Additionally, the system was grounded in INASL clinical guidelines, ensuring evidence-based and consistent treatment recommendations while reducing hallucination risks.

The platform was further developed into a clinically deployable system, incorporating secure authentication (OAuth 2.0), role-based access control, audit logging, and encrypted data handling to support real-world usage. A structured evaluation framework inspired by recent medical AI benchmarks was designed to assess calculation accuracy, reasoning consistency, hallucination rates, and overall report quality.

Preliminary validation on real-world de-identified HCC patient cases demonstrated approximately 90% agreement with clinician decisions, indicating strong alignment with expert judgement. Observed discrepancies primarily arose from non-clinical factors such as financial constraints and patient-specific considerations not captured in structured data. While large-scale quantitative evaluation remains ongoing due to limited annotated datasets, the system demonstrates promising feasibility as a scalable, explainable, and clinically grounded decision support tool.

By integrating multimodal data processing, guideline-aligned reasoning, and clinician-in-the-loop validation within a secure and deployable framework, Samavet AI represents a significant step toward democratizing access to multidisciplinary oncology expertise and improving consistency in cancer care delivery.

Keywords: Tumour Board, Multimodal AI, Clinical Decision Support, Medical Agents, Radiology AI, HCC, Explainable AI, Knowledge Graphs, LLM-based Reasoning

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Work Timeline

The progress of this project took place between August 2025 to April 2026. The work was carried out in iterative phases, evolving from system design to deployment and clinical validation.

- **August 2025**
  - Conducted literature review on tumour boards, multimodal AI, clinical agent architectures, and HCC treatment workflows.
  - Analyzed limitations of existing clinical decision support systems and tumour board automation approaches.
  - Designed the initial Samavet AI system architecture and agent-based processing framework.
  - Initiated collaboration with clinical advisors and aligned project scope with real-world tumour board workflows.
- **September 2025**
  - Developed the Samavet AI dashboard (React frontend + Python backend) for patient entity creation and management.
  - Designed a structured patient schema supporting multimodal data ingestion (labs, imaging, clinical notes, timelines).
  - Implemented plug-and-play agent integration using JSON/Protobuf-based routing formats.
  - Defined system workflow, architecture diagrams, and UI/UX structure.
- **October 2025**
  - Developed core AI agents, including:
    * Clinical Summarisation Agent (LLM-based synthesis of text and tabular data)
    * Imaging Interpretation Agent (CT/MRI feature extraction and lesion characterization)
    * Strategy Planning Agent (tumour-board style reasoning)
  - Built orchestration and communication layer for multimodal reasoning.
  - Designed initial knowledge graph structures and explainability components.
  - Performed feasibility testing using synthetic and anonymised datasets.

- **November 2025**
  - Integrated backend pipeline with dashboard UI for end-to-end execution.
  - Enabled ingestion of anonymised retrospective patient datasets.
  - Defined evaluation methodology and system objectives.
  - Incorporated early clinical feedback into system refinements.
  - Prepared initial research documentation (abstract, background, problem formulation).
- **December 2025 – January 2026**
  - Integrated agent pipeline with frontend dashboard for real-time interaction and visualization.
  - Developed patient list dashboard with search, filtering, sorting, and pagination capabilities.
  - Implemented patient view interface with modular agent outputs and consolidated tumour board summaries.
  - Ensured robust handling of partial agent failures, API latency, and incomplete data scenarios.
  - Implemented secure authentication using Google OAuth 2.0 and JWT-based session management.
  - Developed role-based access control, master dashboard, and audit logging for system governance.
- **February 2026**
  - Deployed the system on institutional servers with secure external access for clinical collaborators.
  - Implemented encryption and secure storage mechanisms for patient data.
  - Enhanced schema design with HCC-specific clinical requirements and decision-aware validation.
  - Developed PDF-based ingestion pipeline for auto-populating structured patient data.
- **March 2026**
  - Introduced a knowledge graph–based reasoning engine using ReAct-style planning.
  - Implemented rule-based decision routing for efficient and context-aware agent invocation.
  - Grounded tumour board recommendations in INASL clinical guidelines to ensure evidence-based outputs.
  - Optimized agent execution by caching responses and reducing redundant LLM calls.
  - Modeled clinician workflows and incorporated decision logic into system design.
- **April 2026**
  - Conducted preliminary validation on real-world de-identified patient cases.
  - Compared system outputs with clinician decisions, observing strong alignment.
  - Designed a comprehensive evaluation framework for future large-scale validation.
  - Finalized system for demonstration and report submission.

Contents

Chapter 1

Introduction

1.1 Tumour Board

A tumour board, also referred to as a multidisciplinary team (MDT) meeting, is a structured clinical decision forum in which specialists from medical oncology, radiology, pathology, surgery, and radiation oncology collaborate to review patient information and collectively recommend an evidence-based treatment plan (National Cancer Institute, 2024). Tumour boards serve as a critical component of modern oncology practice by ensuring that complex cancer cases are evaluated from multiple clinical perspectives rather than relying on an individual clinician's interpretation (Barnes-Jewish Hospital, 2023).

Tumour board meetings should ideally occur on a weekly basis in established cancer centres, where new and ongoing cases are presented and discussed to determine disease staging, treatment pathways, and response evaluation (Bigo Health, 2025). Evidence demonstrates that multidisciplinary review leads to improved staging accuracy, guideline-compliant care, and measurable improvements in survival outcomes for certain cancer types, particularly when cases undergo repeated review (Liu et al., 2020). Tumour boards also improve documentation quality and consistency in treatment selection by promoting structured discussion and shared accountability among specialists (Soukup et al., 2022).

However, despite their clinical value, tumour boards are resource-intensive and demand extensive preparation, coordination, and synchronous participation. In many low-resource or high-workload healthcare environments, limited specialist availability, high patient volume, and administrative overhead result in infrequent or delayed tumour board meetings, sometimes occurring only once in several weeks (Okasako et al., 2022). This reduces timely access to expert consensus and contributes to variability in care delivery. As cancer treatment becomes increasingly data-rich and multimodal, integrating imaging, pathology, molecular markers, and longitudinal clinical history, traditional tumour board workflows face significant scalability challenges. These constraints highlight the need for technology-enabled support systems capable of augmenting multidisciplinary reasoning and improving accessibility to expert-level insights.

1.2 Background and Motivation

The management and treatment of cancer require interpretation of complex, heterogeneous medical information including laboratory trends, pathology reports, free-text clinical notes, radiology imaging, and longitudinal treatment responses. The cognitive and operational workload associated with manually aggregating, summarising, and presenting these multimodal data sources places significant pressure on physicians preparing for tumour board discussions. This challenge is amplified in environments with limited specialist staffing or inadequate digital infrastructure, where tumour board meetings may not occur frequently enough to support timely decision-making and treatment planning.

While AI-based decision support tools have recently emerged across diagnostic and imaging domains, most existing solutions operate in single-modality silos, lack reasoning transparency, or do not support the multidisciplinary workflow required to represent tumour-board-style decision making. Furthermore, there is limited technology that enables the automated synthesis of multimodal cancer data into interpretable, structured clinical summaries that can support clinician review. These gaps restrict the scalability of high-quality cancer care, particularly in hospitals that cannot consistently host full multidisciplinary tumour board meetings.

The motivation behind Samavet AI is to address these limitations by building a scalable tumour-board emulation system capable of integrating multimodal patient data and generating structured, explainable clinical insights. In the initial phase of this work, the focus was on designing the system architecture, developing a structured patient schema, and implementing a modular agent-based pipeline for multimodal reasoning.

In the extended phase presented in this report, the system has been significantly enhanced into a **clinically deployable and reasoning-aware platform**. This includes the development of a secure web-based dashboard with authentication and access control, integration of knowledge graph–based reasoning for context-aware decision making, grounding of outputs in clinical guidelines, and deployment on institutional infrastructure for real-world usage. Additionally, preliminary validation against clinician decisions has been conducted to assess system alignment with expert judgement.

By reducing dependence on synchronous specialist participation and providing consistent, evidence-derived reasoning assistance, Samavet AI aims to democratise access to multidisciplinary oncology insight and improve equity, efficiency, and transparency in cancer decision support.

Chapter 2

Related Work

2.1 GPT for Oncology

Recent advances in AI have yielded powerful autonomous agents for clinical decision support and biomedical research. For example, Ferber et al. (2025) developed an autonomous oncology decision-support agent that integrates GPT-4 reasoning with specialized medical tools. Their system uses vision transformer mocentredanalyze histopathology slides (e.g. detecting MSI, KRAS/BRAF mutations) and MedSAM for radiology segmentation, alongside web search of guidelines (OncoKB, PubMed, etc.). In a benchmark of 20 multimodal cancer cases, the AI agent selected appropriate tools (87.5% accuracy) and reached correct clinical conclusions in 91.0% of cases. Critically, Ferber et al. report that this integrated agent "drastically improved decision-making accuracy from 30.3% to 87.2%" compared to GPT-4 alone. Their results illustrate how coupling an LLM with domain-specific models and data sources can greatly enhance oncology decision accuracy, laying a foundation for AI-assisted tumor-board reasoning.

2.2 LLM as Clinician Copilot

OpenAI's recent work with Penda Health demonstrates an LLM "clinician copilot" for real-world clinical decision support. Chen et al. (2025) studied AI Consult, an LLM-based safety-net tool embedded in a Nairobi primary care EMR. In a pragmatic trial over 39,849 patient visits, clinicians using AI Consult had significantly fewer clinical errors: e.g. history-taking errors fell by 32%, investigation errors by 10%, diagnostic errors by 16%, and treatment errors by 13% relative to a control group. Surveyed clinicians overwhelmingly found the tool helpful, with 75% reporting substantially improved care. The study attributes this success to three factors: a highly capable LLM model, a clinically aligned implementation (co-designed with users), and active deployment strategies. Notably, AI Consult uses a traffic-light alert interface (green/yellow/red) to flag actionable advice at key workflow points, minimizing disruption. This work exemplifies how an LLM-based assistant, when thoughtfully integrated into clinical workflow, can measurably improve care quality.

2.3 Biomedical AI Agent

General-purpose biomedical AI agents have also been proposed. Huang et al. (2025) introduced Biomni, a "generalist biomedical AI agent" designed to autonomously execute a wide range of research tasks. Biomni is built on a unified biomedical action space that includes a curated library of 150 specialized tools, 105 software packages, and 59 databases. The agent architecture flexibly composes these tools via LLM-driven planning and code execution, enabling it to formulate hypotheses, perform large-scale bioinformatics analyses, and design experiments across diverse subfields. Rigorous benchmarks and case studies demonstrated Biomni's strong performance on varied biomedical Q&A tasks and novel problem scenarios. Similarly, Agent Laboratory (Schmidgall et al., 2025) is a multi-stage LLM framework for accelerating scientific research. Agent Lab takes a human research idea and automatically executes literature review, experimentation, and report-writing via a pipeline of specialized LLM "agents". In evaluations, Agent Lab produced publishable research reports and code, showing that an orchestrated multi-agent system can largely automate end-to-end discovery workflows. Together, these systems illustrate how modular AI agents can be composed to tackle complex biomedical tasks without manual intervention.

2.4 Clinical LLMs

Another line of work focuses on domain-specific clinical LLMs. For example, Simsek et al. (2025) developed GastroGPT, a gastroenterology-specialized LLM trained on GI clinical data. In a structured evaluation across seven gastroenterology tasks, GastroGPT significantly outperformed general-purpose LLMs (GPT-4, Bard, Claude). It achieved a higher overall score (8.1/10) than GPT-4 (5.2/10) and others, outperforming them on six out of seven tasks. Its performance was robust across case complexities and it exhibited greater consistency. The authors conclude that a specialty-focused clinical model can yield "superior utility" on relevant tasks compared to general models. This work suggests that tailoring LLM capabilities to a medical subdomain (e.g. GI oncology) can boost reliability and relevance.

2.5 Personal Health Agent - Multi Agent Orchestration

Finally, Google's Personal Health Agent (Heydari et al., 2025) illustrates a multi-agent framework for personal healthcare. Heydari and colleagues performed user-centered analysis to identify consumer health needs and then built PHA, a multi-agent system with three specialized sub-agents. The agents include a data science agent (analyzing wearable time-series and health records), a health expert agent (integrating personal context for insights), and a health coach agent (providing personalized advice and tracking progress). The PHA system dynamically orchestrates these agents to address individual goals. In extensive automated and human evaluations on benchmark tasks, PHA demonstrated comprehensive capabilities in reasoning over multimodal personal health data, representing one of the most thorough health-agent assessments to date. Although PHA targets consumer wellness rather than clinical care, it exemplifies how diverse data and multi-agent architectures can be combined for personalized health assistance.

2.6 Caveat

Collectively, these prior works showcase key innovations: integration of LLMs with specialized medical tools (Ferber et al.), real-world clinical co-pilots (Penda/Chen et al.), orchestration of multi-agent pipelines (Agent Laboratory, Biomni, PHA), and domain-specific LLM design (GastroGPT). They highlight strategies for improving accuracy, explainability, and user alignment in health AI. However, none directly addresses the tumor board setting, which requires emulating a multidisciplinary, case-based cancer review.

Chapter 3

Research Context and Components

3.1 Premise

The related work above motivates a system that integrates multimodal AI into the complex workflow of oncology tumor boards. Tumor boards require consolidating diverse patient data (imaging, pathology, labs, notes) and supporting multi-specialist decision-making. Drawing on Ferber et al.'s demonstration of multimodal oncology reasoning and the user-centered deployment strategies of Penda Health's LLM copilot, our project aims to develop Samavet AI, a tumor-board emulation platform aligned with clinical workflows.

In the initial phase, the focus was on designing a modular agent-based architecture and structured data representation. In the extended phase presented in this report, this design has been realized into a clinically deployable system with secure data handling, workflow-aligned interfaces, and reasoning-aware agent orchestration.

In particular, Samavet AI combines a structured patient data dashboard, flexible agent orchestration, knowledge-guided reasoning, and explainable multimodal analysis to fill a critical gap: existing AI tools address individual diagnostic tasks or research workflows, but none offers an end-to-end, interactive tumor-board assistant that reflects real-world oncology decision-making.

3.2 Key Components and Innovations

Samavet AI introduces several key components and innovations:

- Dashboard with structured patient data management: A secure web-based dashboard presents each patient's case information in a unified view. Patient entities (demographics, history, lab results, imaging findings, pathology, etc.) are organized according to a predefined schema (illustrated by our JSON/Protobuf case definition). This structured management ensures that all relevant data are accessible and annotated. Clinicians can review and update entities, while the system's agents consume the same structured representation. The dashboard also surfaces AI findings with clear highlights, e.g. showing extracted summaries from notes or flagged lab values, and visual indicators of uncertainty or confidence. By giving an overview of the case in a familiar format, the dashboard emulates the information flow of an actual tumor board while ensuring data coherence.
- Modular plug-and-play agent orchestration: Underlying the dashboard is a modular AI agent framework inspired by generalist agents. Each specialized sub-agent performs a

specific task: for example, one agent generates a brief summary of clinical notes, another analyzes imaging scans (using integrated vision models), another retrieves relevant literature or guidelines, and yet another proposes treatment options based on protocol. These agents can be dynamically composed: the orchestration engine sequences agents as needed (e.g. first extract history, then perform image segmentation, then run drug-interaction check) and integrates their outputs. New agents (or updated ones) can be added by simply defining their inputs/outputs, enabling a plug-and-play architecture similar to Biomni's extensible toolset. In effect, Samavet's orchestration borrows the multi-agent pipeline ideas of Agent Laboratory and the data-driven sub-agent model of Google's PHA, but applies them to tumor-board tasks.

- Data integration via JSON/Protobuf schemas: To handle heterogeneous clinical data, Samavet uses structured data formats. Patient records (from EMR exports, PACS images, lab databases, etc.) are ingested and serialized into a unified JSON/Protobuf format. This standardized schema ensures that both the UI and agents access data consistently (similar in spirit to FHIR-based efforts). By defining clear types and examples for each field (e.g. in our JSON schema), we enable type-safe data exchange. Agents read from and write to these structured objects, facilitating collaboration between text- and code-based components. The schema-driven design parallels how large biomedical agent systems catalog tools and data resources (e.g. Biomni's action space), but here focuses specifically on patient-centric data for oncology cases.

- Multimodal processing pipeline: Samavet implements a multimodal AI pipeline for cancer cases. Textual components (doctor's notes, pathology reports) are summarized by an LLM agent that highlights key findings. Medical images (CT/MRI scans, histology slides) are processed by deep-learning vision models: for example, an MRI image analysis agent segments tumor volumes and measures features (inspired by Ferber's MedSAM use). A reasoning agent then integrates all modalities: it considers the extracted text facts, imaging findings, lab values, and known oncological knowledge to infer stage, prognosis, and treatment options. Under the hood, retrieval-augmented methods fetch relevant guidelines (e.g. NCCN recommendations) or research articles when needed. Finally, an LLM synthesizes a coherent case summary and proposed plan. This multimodal fusion goes beyond single-mode AI; it builds on Ferber et al.'s finding that combining LLMs with vision and search tools substantially improves accuracy, applying it end-to-end for whole-case reasoning.

- Explainable outputs and confidence highlighting: A central design goal is transparency. Each agent's output includes explanations or evidence. For example, the summary agent not only states conclusions but cites specific report snippets. The recommendations module indicates its confidence level, and the UI uses color/highlight cues (inspired by Penda's traffic-light alerts) to draw attention to uncertainties or critical alerts. We also support a knowledge-graph style view: extracted clinical entities and their relationships (e.g. symptom→diagnosis→treatment) are displayed so clinicians can trace the reasoning. These features aim to build trust by making the AI's "chain of thought" inspectable. This follows prior work showing that knowledge-graph reasoning and clear signal cues improve interpretability in oncology AI and aligns with Penda's emphasis on explainability in the UI.

- Knowledge graph–based reasoning and decision routing: In the extended phase, a reasoning layer was introduced that converts structured patient data into a knowledge graph representation. Using ReAct-style planning and rule-based logic, the system dynamically determines which agents to invoke based on clinical context. This improves both computational efficiency and interpretability by aligning agent execution with clinical decision pathways.

- Guideline-grounded clinical reasoning: The tumor board agent is grounded in established clinical guidelines (e.g., INASL), ensuring that generated recommendations are evidence-based and consistent with standard oncology practices. This significantly reduces hallucination risks and improves trustworthiness of outputs.

- Clinical workflow alignment: The system design incorporates detailed modeling of clinician workflows, including diagnostic sequencing, specialist interactions, and treatment decision hierarchies. This ensures that agent orchestration and output structure closely mirror real tumor board discussions, improving usability and adoption potential.

3.3 Unique Value Proposition

Samavet AI builds on these prior systems by leveraging their strengths. It is inspired by the multimodal precision of Ferber et al. (image and histology analysis), the clinically-aligned deployment strategy of Penda (user-centered design and alerting), the modular tool integration of Biomni and Agent Lab, and the task-specific focus of models like GastroGPT.

Additionally, Samavet AI is explicitly tailored to emulate multidisciplinary tumor boards not only in architecture but also in workflow, reasoning, and deployment context. Unlike prior systems that focus on isolated tasks, Samavet AI integrates structured patient data, multi-specialist reasoning, and guideline-based decision making into a single platform.

A key distinguishing aspect of this work is the transition from a conceptual architecture to a deployed, secure, and clinically accessible system, incorporating knowledge graph–based reasoning, workflow-aware orchestration, and preliminary validation against clinician decisions. By aligning with real-world constraints and clinical practices, Samavet AI provides a practical and scalable approach to AI-assisted oncology decision support.

Chapter 4

Patient Schema

4.1 Overview

We have designed the patient schema as a structured, clinically grounded representation that encodes all relevant information required for tumour board–style reasoning. The schema serves as a unified interface between clinicians, data ingestion pipelines, and AI agents, enabling consistent and interpretable decision-making across the system.

In the initial phase of this work, the schema focused on organizing multimodal patient data (clinical, laboratory, imaging, pathology, and treatment history) into a structured JSON format for downstream agent consumption. In the extended phase, the schema was significantly enhanced to support real-world clinical workflows, incomplete data scenarios, and agent-based reasoning.

A key design principle is the explicit distinction between **"not assessed"** and **"measured values"**. Fields that are not documented in clinical reports are stored as `null`, whereas explicitly observed values (including "none" or zero) are recorded as valid entries. This distinction is critical for clinical reasoning, as absence of evidence is not equivalent to evidence of absence, particularly in staging systems such as BCLC.

The schema is implemented in JSON, enabling seamless integration with web-based dashboards, API pipelines, and AI agents. It follows a modular and hierarchical structure, allowing flexible inclusion or omission of sections depending on data availability. This design aligns with healthcare interoperability standards such as HL7 FHIR and OMOP-CDM while remaining lightweight and extensible for research and deployment contexts.

Furthermore, the schema is tightly coupled with the system's agent architecture. The structured patient input schema is transformed into intermediate agent-specific representations, and the outputs of modular agents are serialized into a standardized format that serves as input to the tumour board orchestration layer. This end-to-end schema design enables reproducibility, traceability, and efficient multi-agent collaboration.

Category	Description	Data Type / Notes
Case ID	Unique identifier for each patient case	String (auto-generated or user-defined)
Demographics	Age, sex, BMI, identifiers	Numeric, Enum
Clinical Data	Etiology, symptoms, comorbidities, ECOG, ascites, encephalopathy, notes	Structured text, Enums, Null-aware
Lab Data	Baseline labs, longitudinal labs, derived scores (Child-Pugh, MELD, ALBI)	Numeric, Time-series, Computed fields
Radiology	Imaging studies, lesion-level annotations (size, LI-RADS), PVTT, metastasis	Nested objects, Arrays
Pathology	Biopsy findings, tumor grade, vascular invasion, molecular markers	Boolean, Enum, Text
Treatment History	Previous and current treatments, response	Structured arrays, Text
Tumour Board Data	Multidisciplinary notes, members present	Text, Arrays
Ground Truth	Clinical scores, BCLC stage, treatment intent	Enums, Numeric, Evaluation targets
Data Completeness	Missing vs assessed distinction	Null vs explicit value semantics

Table 4.1: Enhanced Patient Input Schema for Samavet AI

4.2 Components:

4.2.1 Demographics

The Demographics section captures basic patient identifiers and attributes critical for context and risk stratification. Typical fields include patient age (numeric), biological sex (enumerated string "M" — "F"), and region or ethnicity (string or code). These fields are mandatory to describe the subject cohort. Age and sex strongly influence disease risk and treatment guidelines, so the schema uses a number type for age (years) and an enum for sex. We omit directly identifiable fields (name, ID) or replace them with anonymized keys to comply with privacy; instead, an internal "case id" ensures record linkage. Using primitive types here ensures easy validation and standardized encoding (e.g. LOINC/HL7 value sets for sex). Downstream, agents use demographics for filtering cohorts, risk estimation, and bias checks. For example, an AI oncologist agent may adjust baseline prognosis based on age/sex. Demographics also enable integration with external systems: many EHR/CDM schemas (OMOP Patient table, FHIR Patient resource) include matching fields, easing data import/export.

4.2.2 Clinical Presentation

The Clinical Presentation (or summary) section describes how the disease manifested. It includes chief complaint, underlying etiology, symptom list, and comorbidities. We represent this as a JSON object with semantic keys, combining strings and lists

The choice of an object (rather than a free-text blob) ensures structured data extraction: "etiology" is a string (often categorical code or controlled term); "symptoms" and "comorbidities" are arrays of strings or codes. Enums or coded values (ICD/SNOMED where possible) can further standardize entries. This structure was shaped by clinical input: during schema reviews, clinicians stressed capturing etiology (viral vs. alcoholic), which guides staging and treatment. Arrays were chosen for symptoms/comorbidities because multiple items may apply; it also allows AI agents to iterate and query each condition. In analysis, these fields feed NLP modules (e.g. embedding symptom lists) and rule-based logic (e.g. "if cirrhosis+portal hypertension, flag eligibility for transplant"). Standardizing them (rather than free text) directly supports downstream reasoning. In practice, the "clinical_presentation" object ensures all agents see the same key-value patterns and controlled values.

4.2.3 Lab Data

The Lab Data section records numeric measurements, divided into baseline values and longitudinal follow-ups. Here, "baseline" is an object of key-value pairs (lab name to value) for admission labs. The "follow_up" is an array of timestamped lab records (each an object with date and values), capturing trends. We chose arrays of objects for time series to accommodate variable numbers of visits and facilitate charting/analysis. Numeric types are used for continuous labs, and strings/enums for categorical scores. The "derived_scores" object holds calculated indices (Child-Pugh, MELD-Na, AFP, etc.) as either numeric or coded values. These are crucial ground truth indicators of severity. This section reflects feedback from hepatologists who emphasized tracking liver function over time. In downstream workflows, trend data allow agents to detect improving or worsening patterns (e.g. rising AFP levels). Structured lab fields (rather than embedding in text) enable quick queries (e.g. "list all AST values") and compatibility with analytics (e.g. plotting trajectories). The schema aligns with standards by using standard lab codes (LOINC) where possible, and by representing units and dates explicitly. Overall, this design supports clinical monitoring and AI modeling of disease progression.

4.2.4 Radiology Reports and Imaging Files

The Radiology section encodes findings from imaging studies and links to raw image data. It includes fields such as modality (enum "CT"/"MRI"), report impressions, lesion annotations, and imaging references. Here, "findings" is an array of objects (one per scan timepoint or lesion group). Each object uses specific data types: dates (ISO strings), counts/measurements (numbers), and coded fields (e.g. LIRADS as integer or enum, PVTT as boolean). This design reflects radiologist-annotated report data. Attaching file URIs under "attachments" (array of strings) points to the original DICOM images for agent retrieval. Using arrays permits multiple imaging timepoints or multiple lesion records per study. The schema also reserves space for structured radiology scores (LIRADS, RECIST) because these are critical outcome metrics. This structured approach follows radiology reporting trends: for example, structured templates have been shown to improve clarity and completeness of findings. By encoding image references as JSON paths, AI vision agents can fetch and process scans. The modular object also separates raw report text (if captured) from parsed fields; agents can generate text from fields or vice versa. In summary, the radiology section provides a clear, structured summary of imaging, enabling downstream analytics (trend plotting, lesion tracking) and integration with image-analysis modules. Standards such as DICOM (for images) and LOINC codes (for observations) are implied here, facilitating interoperability.

4.2.5 Pathology

The Pathology section summarizes histopathological findings from biopsies or resections. Key fields include tumor type, grade/differentiation, margin status, and molecular markers. Each sub-field is typed for precision: diagnostic terms as strings or coded concepts, booleans for presence/absence (vascular invasion), and arrays for marker lists. We chose this object structure after consulting pathologists, who requested discrete capture of tumor characteristics. The format reflects synoptic reporting standards: structured pathology templates markedly outperform narrative reports in completeness, so our schema enforces key data elements (grade, margins, invasions) explicitly. Using nested objects (e.g. "ihc_markers") allows expansion for multiple markers without ambiguity. Downstream, these fields feed classification agents and verify imaging-based predictions. For instance, if an AI agent suggests a diagnosis, it can be checked against biopsy_diagnosis. Molecular and IHC data support precision medicine workflows. Importantly, storing pathology in structured form facilitates linking to outcomes: final pathology stage or recurrence. If molecular data grow (e.g. next-gen sequencing), new fields can plug into this section. Overall, the pathology schema ensures critical tissue-derived truths are captured for AI analysis and validation.

4.2.6 Treatment History

The Treatment History section records all prior therapies (locoregional or systemic) the patient received, in chronological order. Each entry includes therapy type, date, and relevant details. This is represented as an array of objects, one per intervention. We use dates (ISO format) and strings for type (with controlled vocabulary: e.g. "TACE", "RFA", "Chemotherapy") to ensure consistency. Numerical fields capture doses or cycles if needed. This design allows flexible extension: new treatment modalities can be added as new entries or fields. Clinicians on our team insisted on detail (e.g. therapy response) to inform future decisions. The structured list makes it easy for an AI agent to review prior treatments (e.g. "Has patient had sorafenib?") and avoid repeats. It also supports generating timelines of therapy. Since different agents might predict or suggest treatments, having a canonical record prevents re-suggesting exhausted therapies. In addition, linking treatment dates with lab or imaging follow-ups (via matching dates) allows evaluation of response. Thus, this modular array contributes to a comprehensive patient timeline, aiding both human and AI decision-support.

4.2.7 Tumour Board Notes

The Tumor Board Notes capture multidisciplinary discussion and recommendations from the tumor board. This is typically a narrative summary (string) or list of bullet points. In our schema, we use a text field (possibly an array of strings) for the final board decision. The format is unstructured text because the content is free-form discussion. We could alternatively structure subfields (opinions, consensus), but opted for simplicity: free text ensures clinicians can articulate nuances. This narrative is important ground truth for evaluating AI recommendations (we compare AI-suggested plan vs. the board's actual plan). In analysis, text notes can be ingested by summarization or comparison agents. Their presence helps train LLMs to mimic human reasoning steps. While not used for quantitative modeling, including the tumor board record ensures the dataset remains true to clinical workflows. For future extensibility, sections or templates could be added here (e.g. separate "recommendations" list), but the current form captures what specialists emphasized in our design discussions. There is no external standard for this free-text narrative, so no citation is needed, but its inclusion reflects the project's focus on emulating the tumor-board process.

4.2.8 Ground Truth Annotations

A Ground Truth section holds validated outcomes and reference standards. This includes clinical staging (e.g. TNM or BCLC stage), final pathology outcome, and imaging annotation truth. In effect, this section contains "answers" against which AI outputs are measured. Fields use enums or standardized codes for staging, and may link to file references (e.g. segmentation masks). Scores are repeated here to ensure consistency with lab entries. By separating ground truth, the schema clarifies evaluation targets (e.g. did the model predict the correct BCLC stage or margin status?). This section was added after clinician feedback on what exactly should be predicted. For example, Dr. Uwais requested a field for "treatment intent" (curative vs. palliative) as a binary target. Recording true imaging annotations (lesion outlines, provided as file paths) supports vision model training. All fields here are of the same basic types (strings, numbers, booleans) chosen for validated outcomes. This dedicated ground-truth block streamlines agent evaluation workflows: comparison scripts can directly read model outputs and compare to schema values.

4.3 Schema Enhancements and System Integration

In the extended phase of this project, the patient schema evolved from a static data representation into a dynamic interface supporting data ingestion, validation, and multi-agent reasoning.

4.3.1 Structured Input Schema for Clinical Data

A comprehensive input schema was defined to capture real-world HCC patient data with high granularity. This includes:

- Explicit value ranges and enums for clinical fields (e.g., ECOG 0–4, ascites severity levels)
- Detailed lab definitions with units, normal ranges, and clinical significance
- Structured radiology representation including lesion-level annotations (size, LI-RADS, PVTT, metastasis)
- Pathology and molecular profiling fields aligned with oncology reporting standards

As shown in the schema reference, each field is typed, validated, and semantically annotated to support both clinical interpretation and machine reasoning.

4.3.2 Handling Incomplete and Real-World Data

A major challenge in clinical data systems is missing or partially documented information. To address this:

- Fields not documented in reports are stored as `null`
- Explicit absence (e.g., no ascites) is encoded as a valid categorical value
- The system flags missing fields required for staging (e.g., ECOG, bilirubin) without blocking data entry

This design ensures that the system remains usable in real-world hospital settings while preserving the integrity of clinical reasoning.

4.3.3 PDF-to-Schema Data Ingestion Pipeline

To reduce manual data entry, a pipeline was developed to convert clinician-generated PDF reports into structured schema-compliant JSON.

As described in the implementation (page 7 of schema reference) :contentReference[oaicite:1]index=1:

- Users upload a PDF via the dashboard
- The system extracts structured data into JSON format
- A validation report is generated to highlight missing or uncertain fields
- Users can edit and approve the structured data before saving

This significantly improves usability and aligns the system with real clinical documentation workflows.

4.3.4 Agent Output Schema and Orchestration Interface

Beyond input representation, the schema also defines the output format of modular agents.

Each agent produces structured outputs (e.g., clinical summary, radiology interpretation, pathology insights) which are serialized into a unified format, as illustrated in the sample output :contentReference[oaicite:2]index=2.

These outputs include:

- Derived clinical scores (Child-Pugh, MELD, ALBI)
- Structured radiology interpretations with lesion-level details
- Pathology summaries and molecular profiles
- Confidence scores and agent metadata

The tumour board orchestrator consumes this standardized output to generate a final consensus report, ensuring modularity and separation of concerns across agents.

4.3.5 Schema as a Reasoning Backbone

The schema is not merely a data container but serves as the backbone for reasoning. Structured fields enable:

- Rule-based decision logic (e.g., BCLC staging conditions)
- Knowledge graph construction from patient attributes
- Context-aware agent invocation based on available data

By enforcing structured representation and semantic clarity, the schema enables reliable and interpretable AI-assisted decision support.

Component	Description	Data Type / Source
Clinical Summary	Aggregated patient condition, symptoms, interpretation	Structured text + JSON
Derived Scores	Child-Pugh, MELD, ALBI with components	Numeric + Nested objects
Lab Interpretation	Flagged abnormalities (high/low/normal)	Key-value pairs
Radiology Summary	Lesion details, LI-RADS classification, PVTT, metastasis	Nested objects, Arrays
Radiology Interpretation	Natural language explanation of imaging findings	Text
Pathology Summary	Biopsy results, differentiation, molecular markers	Structured JSON
Tumour Board Notes	Consolidated multi-specialist reasoning	Text
Treatment Plan Context	Recommended treatments, constraints (e.g., financial)	Structured text
Agent Metadata	Confidence scores, provenance of outputs	Numeric, Metadata fields
Ground Truth (Optional)	Evaluation reference (BCLC stage, intent)	Enum, Numeric

Table 4.2: Modular Agent Output Schema Feeding Tumour Board Orchestrator

4.4 Design Rationale and Clinical Alignment

The final schema reflects extensive clinician input and real-world constraints. We iterated the design in discussions with Dr. Rohit, Dr. Nipun, and Dr. Uwais (PGIMER). Early drafts were broad but unspecific; clinician reviews guided pruning and focus. For example, the team insisted on including specific prognostic scores (Child-Pugh, MELD-Na) and universally recognized staging systems (AJCC TNM, BCLC) so that the data align with standard practice. Oncologists emphasized known risk factors (HBV, comorbidities) and outcomes (recurrence status). Hence, those fields were explicitly added. Radiologists clarified important imaging metrics like LIRADS and mRECIST, leading us to include both numeric and categorical fields in ”imaging”. Pathologists requested structured capture of key histologic features (grade, vascular invasion), which directly influenced the Pathology section design. At each step, clinicians reviewed sample JSONs and adjusted data types (e.g. treating ECOG as integer rather than string). We also aligned terminology with clinical ontology: for instance, treatment types follow NCCN guideline nomenclature, and comorbidities use standard abbreviations (e.g. “DM” for diabetes) per PGIMER practice. Throughout, the feedback loop ensured the schema remained clinically faithful: Dr. Nipun’s team provided example cases to test schema fit, confirming that no critical piece was omitted. The result is a structure that parallels how patient data are documented in practice (as in mCODE's patient and disease profiles) while being optimized for downstream AI use (e.g. enabling agents to query fields deterministically).

4.5 Modularity and Future incorporations

The schema's modular JSON design supports plug-and-play AI agents, time-series analyses, and future extensions. Each top-level section is a self-contained module: an AI agent focusing on labs need only parse "lab_data", while a radiology agent ingests "imaging" and its attachments. Time-series fields (lab follow-ups, imaging follow-ups) are clearly marked as arrays, so analysis pipelines can automatically detect longitudinal data. Because JSON allows undefined fields, the schema can grow without breaking existing workflows. For instance, we can later map these fields onto FHIR or OMOP structures: the modular objects resemble FHIR resources (Patient, Observation, ImagingStudy, etc.) and could be transformed into FHIR-compliant JSON. The design also anticipates ontology tagging: fields like "comorbidities" or "symptoms" could reference SNOMED codes, and stages could link to AJCC codes, promoting machine-readable semantics. As mCODE exemplifies, structured oncology data benefit from controlled vocabularies, and our schema can be extended with value set bindings (e.g. drop-down enums) to adopt ontologies. In practice, this modularity means we can integrate new agents without altering the core schema: a genomic analysis agent could add a "genomics" object alongside existing fields. The plan for future work includes aligning with HL7 FHIR resources and possibly using a JSON Schema or openAPI spec for validation. In sum, the schema's clear hierarchical organization and adherence to interoperability principles ensure it remains extensible and machine-friendly, laying the groundwork for seamless multimodal AI integration.

Chapter 5

System Architecture

5.1 Overview

The Samavet AI Tumor Board Emulation System is designed as a modular, secure, and clinically deployable platform for integrating heterogeneous patient data and facilitating multidisciplinary decision support using AI-driven agents. The architecture comprises a front-end dashboard for clinician interaction, a backend service stack for data processing and orchestration, and a collection of domain-specific AI agents simulating clinical specialties.

In the initial phase, the system focused on designing a modular agent-based pipeline for multimodal reasoning. In the extended phase presented in this report, the architecture has been significantly enhanced to support real-world deployment, secure data handling, knowledge-guided reasoning, and efficient agent execution.

The design follows four primary objectives:

- Enable multimodal patient data integration (text, labs, imaging, pathology)
- Support clinician-guided interaction through a structured and intuitive dashboard
- Orchestrate explainable, guideline-aligned agent-based reasoning pipelines
- Ensure secure, scalable, and efficient deployment for real-world clinical usage

5.2 Frontend Dashboard

The dashboard is developed using React.js with Tailwind CSS for a clean, responsive user interface. It supports dynamic routing, form-based entity creation, data visualization, and AI-agent interactions. The structure supports modular rendering of each schema section.

Key components include:

- Patient Entity Manager: Allows clinicians to create, edit, or delete cases based on a well-defined schema. Supports UUID-based case identification.
- Data Upload Modules: Enables secure upload of DICOM files, pathology PDFs, and JSON schema instances via drag-and-drop or direct input.

- Visualization Engine: Lab and imaging data are plotted using libraries like Recharts, providing time-series graphs for biomarkers such as AFP, bilirubin, or INR.
- AI Interaction Panel: Provides clinicians with real-time access to outputs generated by domain-specific AI agents.

Additionally, the dashboard integrates a PDF-to-structured-data pipeline, allowing clinicians to upload patient reports which are automatically parsed into schema-compliant JSON. A validation interface highlights missing or uncertain fields, enabling users to review and correct extracted data before saving.

The interface also includes dynamic completeness indicators, guiding clinicians on required fields for staging systems such as BCLC and Child-Pugh, thereby improving data quality and usability in real-world settings.

5.3 Backend and API Layer

The backend system is designed in Python (FastAPI), offering a structured RESTful interface for schema validation, agent invocation, and data persistence. The database is managed using PostgreSQL, with support for object storage integration (e.g., S3 for DICOM and PDF references).

Key funcationalities:

- Structured Schema Ingestion: Validates JSON input using custom parsers for each data domain (labs, radiology, clinical notes).
- DICOM and File Handling: Implements preprocessing pipelines for radiological studies, supporting modality and phase detection.
- AI Trigger Handlers: Each AI agent can be invoked via modular endpoints (e.g., /invoke/radiology-agent), receiving patient context in JSON or protocol buffer format.

The backend also incorporates secure authentication and access control mechanisms. Google OAuth 2.0 is used for user authentication, with JWT-based session management ensuring secure communication. Role-based access control (RBAC) differentiates between standard clinical users and administrative users, while audit logs capture system interactions for traceability and compliance.

To support efficient operation, the backend implements caching of agent outputs. Once a patient case is saved, modular agent responses are generated and stored in a structured format. This avoids redundant LLM calls and significantly reduces computational overhead during repeated access or orchestration.

5.4 Agent-Based Clinical Reasoning

The reasoning workflow is built around modular AI agents, each representing a domain-specialist function within the tumor board.

- Strategy Planner: Analyses available data to prioritize modality activation (e.g., labs before imaging)

- Lab Summarizer: Processes numerical and time-series biomarkers, interprets trends, and flags abnormalities
- Radiology Interpreter: Applies models like GPT-V or MedSAM to summarize scan findings in LIRADS or mRECIST terms
- Pathology Agent: Extracts and classifies biopsy report details, including differentiation and vascular invasion
- Specialist Synthesizers: Oncology/Hepatology-like agents that propose recommendations based on domain input
- Orchestration Agent: Gathers sub-agent outputs and creates a coherent, explainable tumor board summary
- Evaluation Agent: Validates coherence, guideline adherence, and identifies hallucinations in AI responses

In the extended system, agent outputs are structured and cached, forming an intermediate representation that is reused by the tumour board orchestrator. Each agent also produces metadata such as confidence scores and provenance, enabling downstream validation and interpretability.

Furthermore, specialist agents are grounded in clinical guidelines (e.g., INASL), ensuring that recommendations remain consistent with established treatment protocols and reducing hallucination risks.

5.5 Knowledge Graph–Based Reasoning Engine

A key enhancement in the extended phase is the introduction of a reasoning layer that transforms structured patient data into a knowledge graph representation. Nodes represent clinical entities (e.g., symptoms, lab values, imaging findings), while edges capture relationships (e.g., disease associations, staging criteria).

The system employs a ReAct-style reasoning framework, combining:

- Structured data extraction from the schema
- Rule-based logic derived from clinical guidelines
- Dynamic agent invocation based on contextual requirements

This enables:

- Context-aware execution of agents (e.g., prioritizing radiology if imaging is critical)
- Reduction in unnecessary LLM calls
- Improved interpretability of decision pathways

The reasoning engine bridges symbolic logic and LLM-based reasoning, allowing the system to operate in a hybrid manner that is both efficient and clinically aligned.

5.6 Integration Pipeline

The system follows a structured end-to-end pipeline:

- **Data Ingestion:** Patient data is entered manually or extracted via PDF parsing into a structured schema.
- **Schema Validation:** Ensures type correctness and flags missing fields required for clinical staging.
- **Agent Execution:** Modular agents process relevant modalities (clinical, lab, radiology, pathology).
- **Intermediate Representation:** Agent outputs are stored in a standardized schema format.
- **Reasoning and Orchestration:** The knowledge graph–based engine determines context-aware agent invocation and aggregates outputs.
- **Tumour Board Composition:** A final structured report is generated.
- **Clinician Review:** Outputs are reviewed, edited, and validated by clinicians.

All stages are logged, traceable, and designed to support human-in-the-loop decision making.

5.7 Tumour Board Composer

This module acts as the final layer of case presentation. It aggregates agent summaries into a human-readable recommendation structured into:

- Clinical Summary
- Diagnostic Findings
- Proposed Treatment Strategy
- Caveats

The composer is grounded in clinical guidelines and incorporates both structured outputs and narrative reasoning, ensuring that recommendations are interpretable, evidence-based, and aligned with real tumour board discussions.

5.8 Deployment and Production Considerations

The system has been deployed on institutional infrastructure with secure external access for clinical collaborators. Deployment considerations include:

- Secure storage and encryption of patient data
- Controlled API exposure for external access

- Scalability for handling multiple concurrent patient cases
- Robust handling of partial failures and incomplete data

These features ensure that the system transitions from a research prototype to a production-ready clinical decision support platform.

5.9 Conclusion

The system architecture of Samavet AI is designed to emulate the layered reasoning process of a real-world tumor board. Its strengths lie in its multimodal flexibility, modular agent ecosystem, and a frontend platform designed for clinical usability. Future extensibility includes integration with EHR standards (e.g., FHIR), federated data pipelines, and voice-to-text interfaces for real-time tumor board meetings.

5.10 System Interface Snapshots

This section presents key interfaces of the Samavet AI platform, demonstrating real-world usability and integration of the proposed architecture.

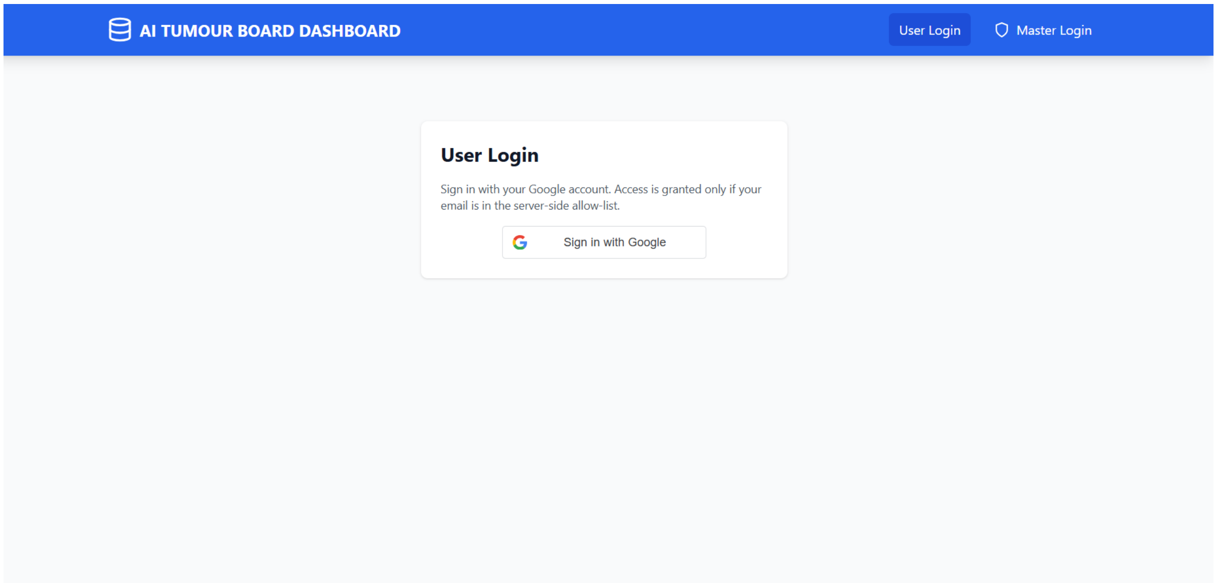

Figure 5.1: User Login Landing page

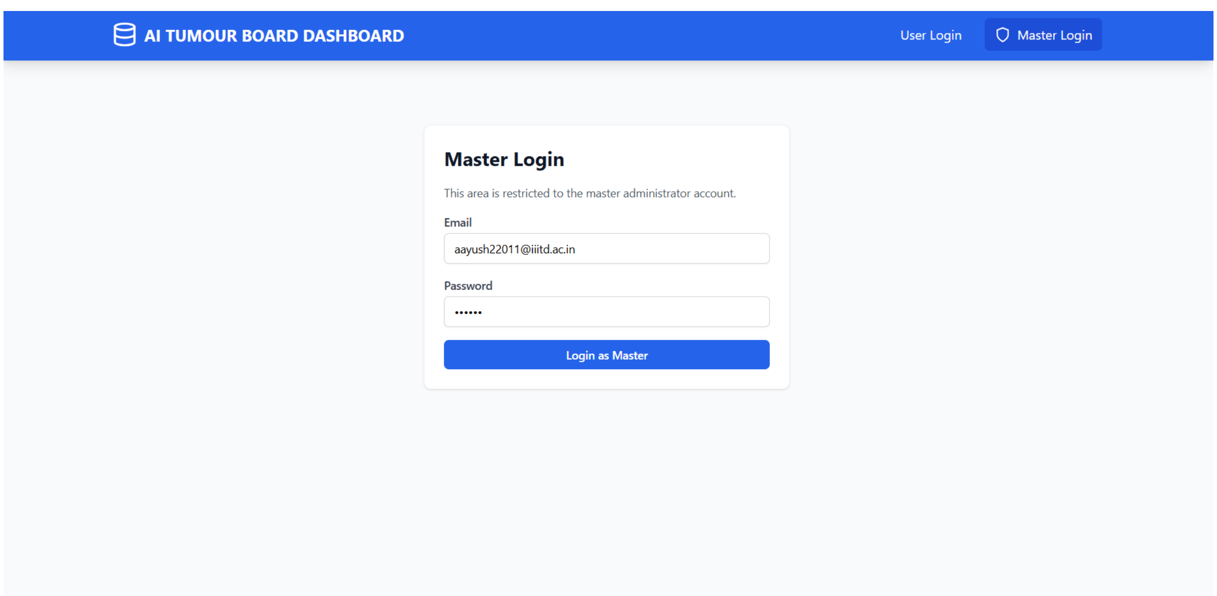

Figure 5.2: Master Admin Login Landing page

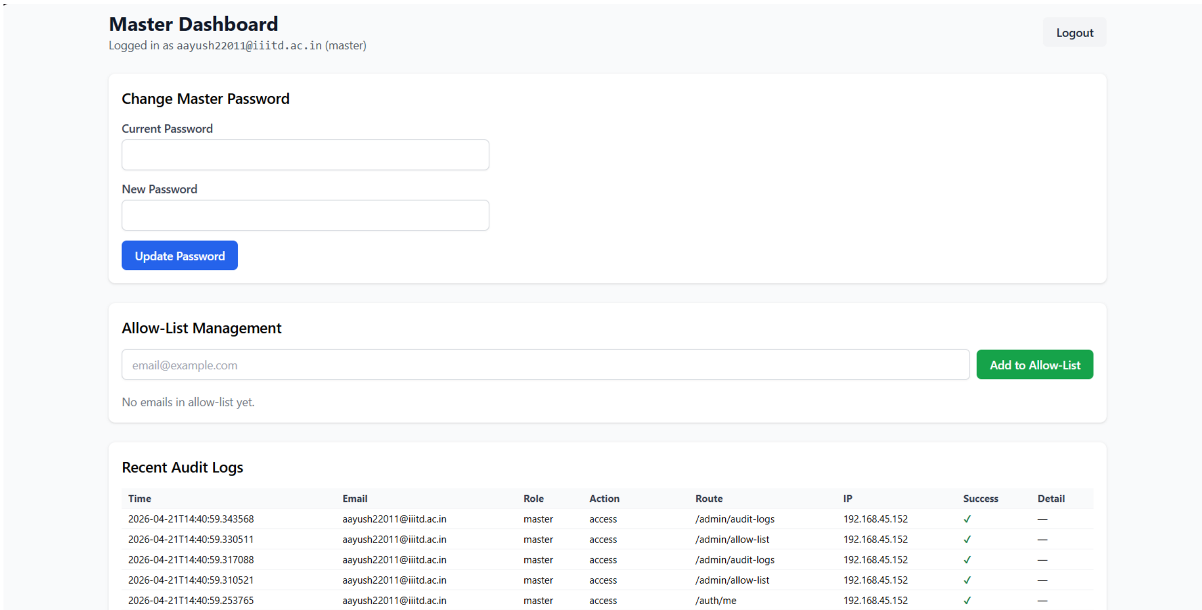

Figure 5.3: Master Admin Dashboard Home Page

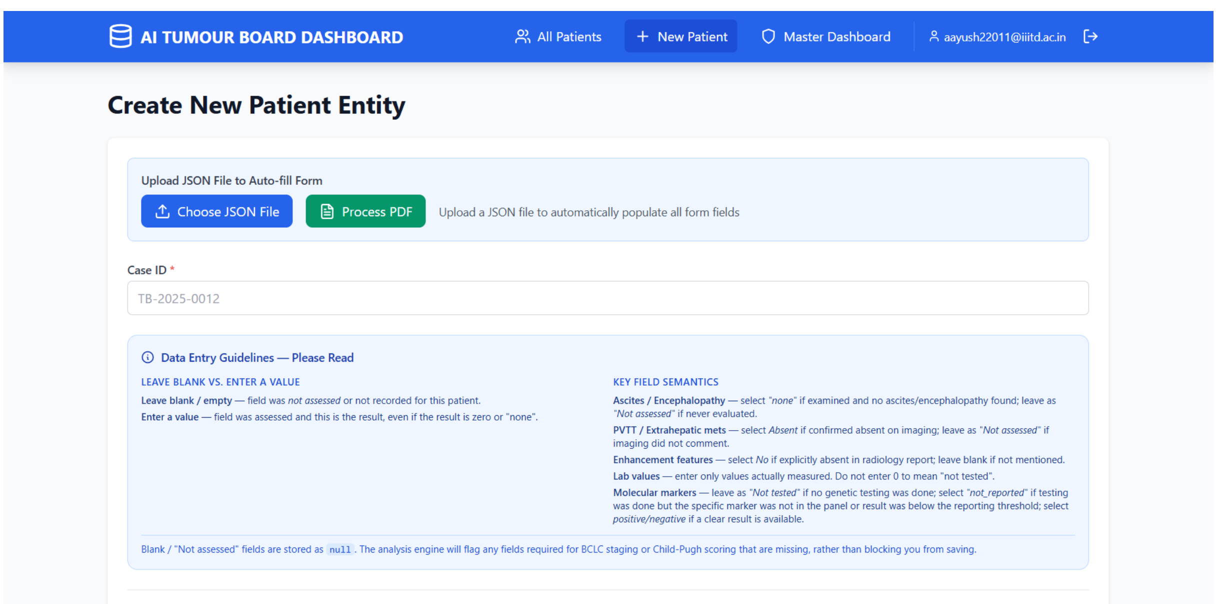

Figure 5.4: New Patient Guidelines

Figure 5.5: Dynamic Checklist while entering Patient data

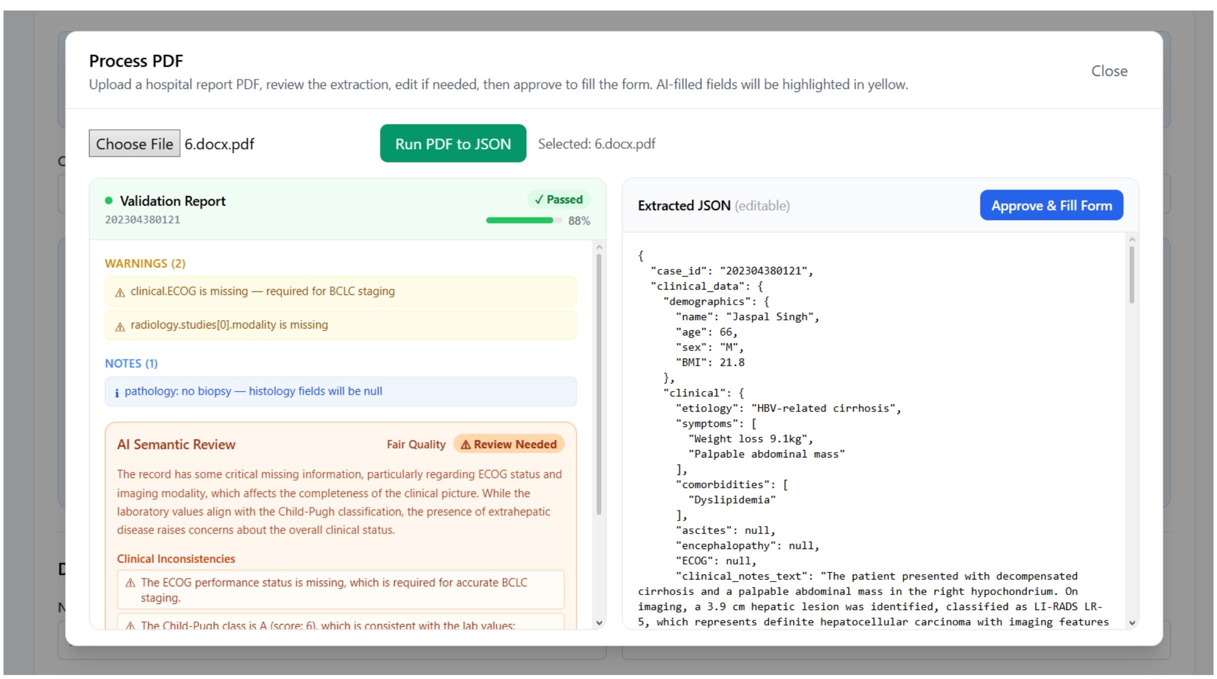

Figure 5.6: Data Validation after uploading pdf

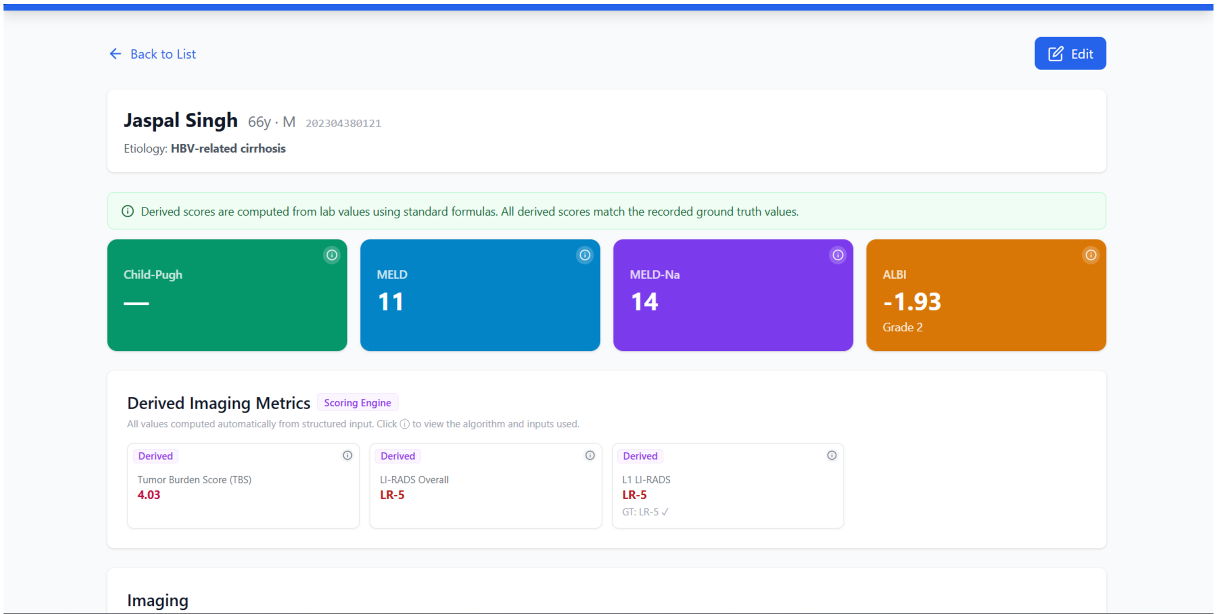

Figure 5.7: View-Patient after saving data

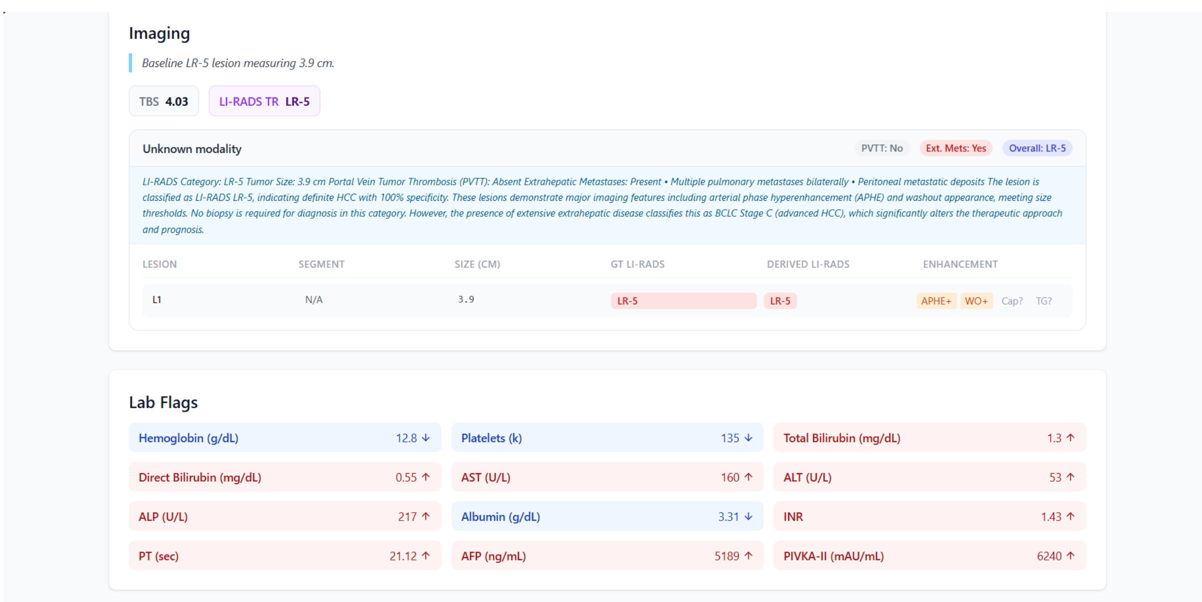

Figure 5.8: View-Patient after saving data

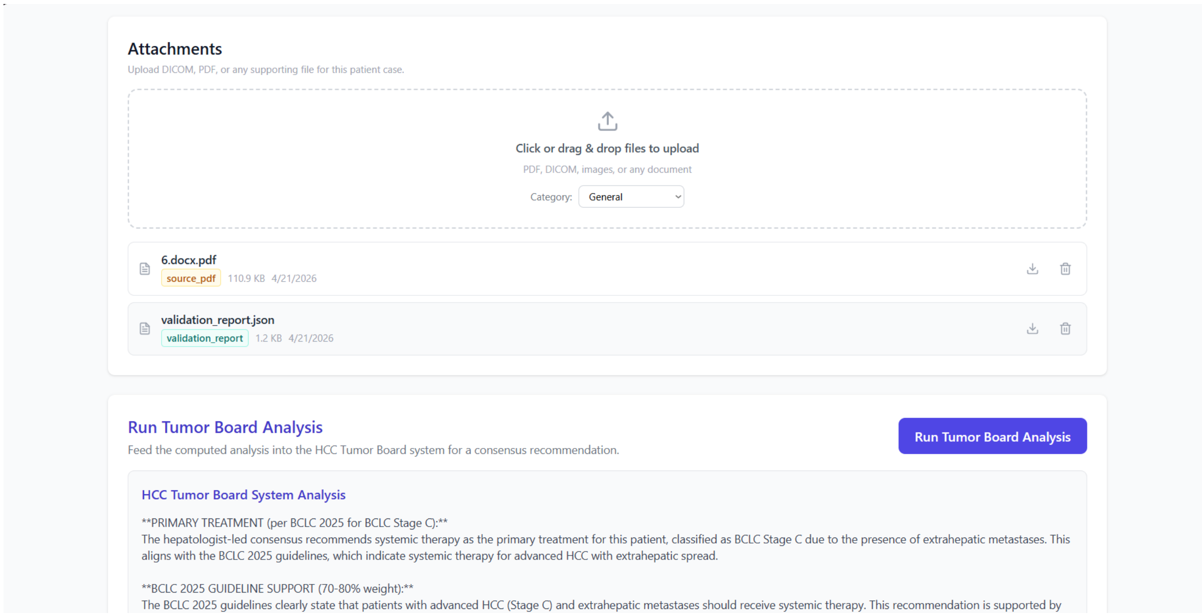

Figure 5.9: Ability to upload documents for more context

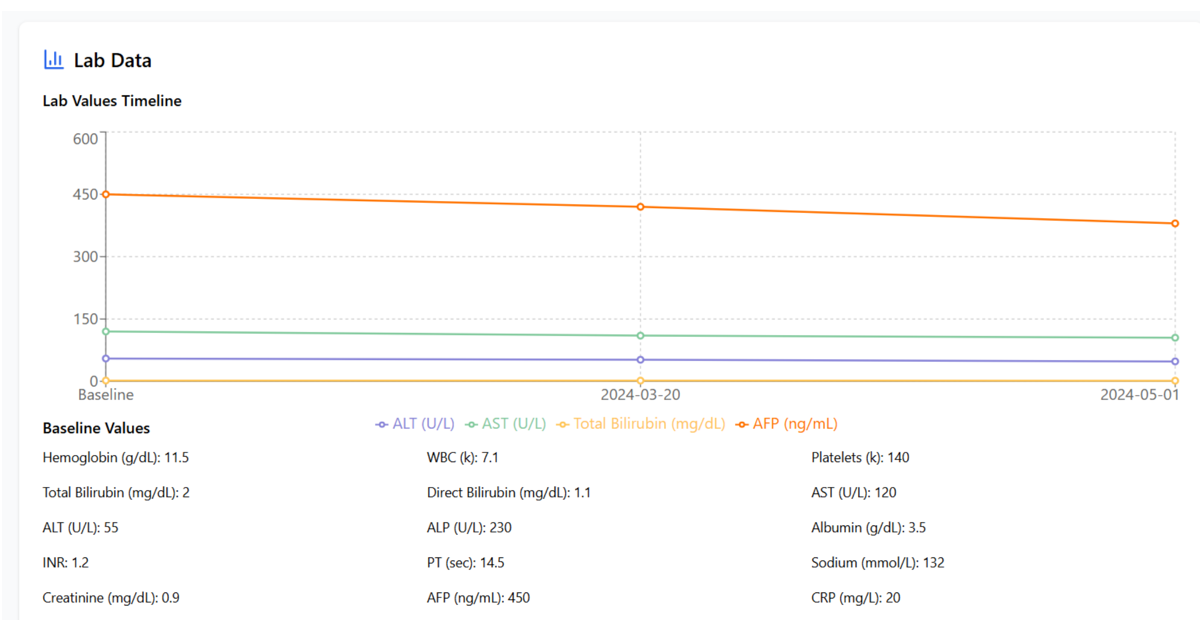

Figure 5.10: lab data illustrated as an interactive chart

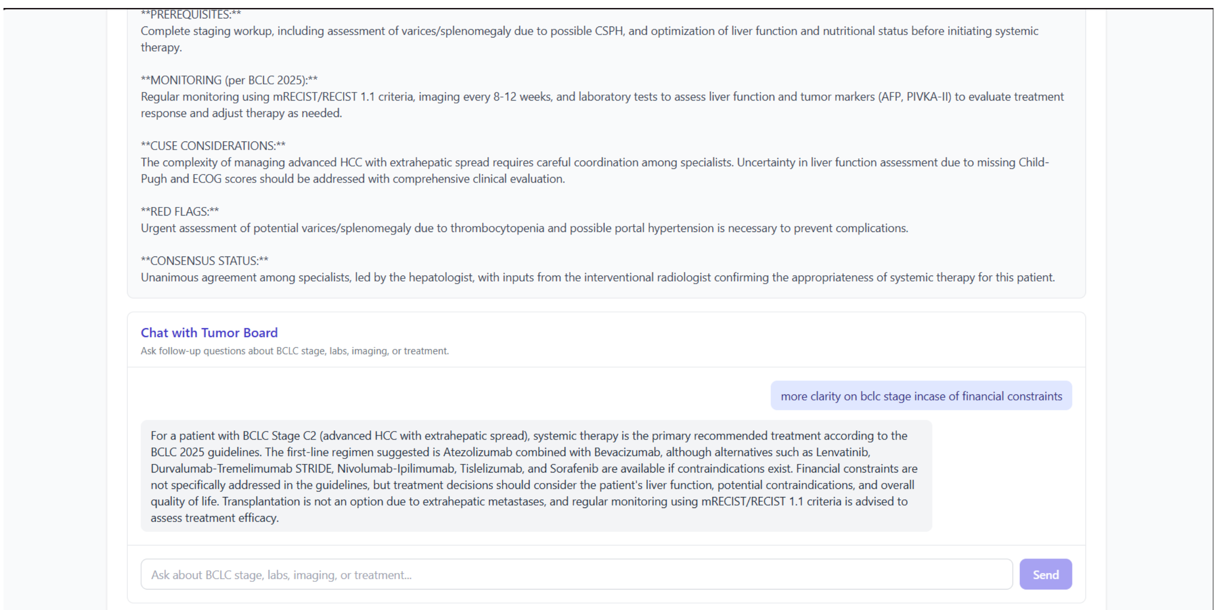

Figure 5.11: Ability to chat with agents (HITL)

Chapter 6

Results and Evaluation

6.1 Deployment and Testing Setup

The Samavet AI Tumor Board Emulation System was deployed on institutional infrastructure and made accessible to clinical collaborators for real-world usage. The system supports secure data entry, PDF-based ingestion, and end-to-end execution of the agent-based reasoning pipeline.

Initial validation was conducted on a set of **four de-identified hepatocellular carcinoma (HCC) patient cases**. Each case was processed through the full pipeline, including structured data ingestion, modular agent execution, and tumour board report generation.

The generated outputs were compared against clinician-provided ground truth decisions derived from multidisciplinary tumor board discussions and established clinical guidelines.

6.2 Observed System Performance

The system demonstrated strong alignment with expert clinical judgement during initial testing.

- **Clinical Agreement:** Approximately **90% agreement** was observed between AI-generated recommendations and clinician decisions across evaluated cases.
- **Strengths:**
  - Accurate computation of clinical scores such as Child-Pugh, MELD, and ALBI.
  - Consistent adherence to guideline-based treatment pathways (aligned with INASL recommendations).
  - Effective synthesis of multimodal inputs including lab data, imaging findings, and clinical notes.
  - Structured and explainable outputs that closely resemble tumour board discussion formats.
- **Explainability:** The system provides traceable reasoning through modular agent outputs, enabling clinicians to understand intermediate decision steps and verify conclusions.

6.3 Case Demonstration

As a representative example, an anonymized HCC patient case was processed through the system, including structured clinical data, imaging findings, pathology details, and treatment history. The system successfully generated a diagnostic summary and treatment plan aligned with tumour board recommendations.

6.4 Limitations Observed During Testing

Despite strong alignment, several limitations were identified during evaluation:

- **Non-clinical factors not captured:** In certain cases, discrepancies arose due to factors not represented in the structured schema, such as:
  - Financial constraints affecting treatment choices
  - Patient preferences or compliance considerations
  - Institutional resource limitations
- **Dependence on input completeness:** Missing key fields (e.g., ECOG, imaging details) can limit the system's ability to fully replicate clinical reasoning, although such cases are flagged by the system.
- **Limited dataset size:** The number of validated cases was limited due to ongoing data collection and dependence on clinician availability for annotation and review.

6.5 Clinical Feedback

Preliminary feedback from clinical collaborators indicated that the system is:

- Useful for structured case summarization and organization of multimodal patient data
- Helpful in generating guideline-consistent treatment suggestions
- Particularly beneficial for assisting junior clinicians in understanding decision pathways

However, broader adoption was constrained by:

- Limited time availability of clinicians
- Ongoing data onboarding processes

6.6 Evaluation Framework

To enable systematic validation, a comprehensive evaluation framework was designed, inspired by recent medical AI benchmarking approaches.

6.6.1 Component-Level Evaluation

- **Calculation Accuracy:** Exact match evaluation for computed clinical scores (Child-Pugh, MELD, ALBI)
- **Structured Output Validation:** Verification of JSON outputs for correctness and logical consistency

6.6.2 Reasoning and Retrieval Evaluation

- **Hallucination Detection:** Ensuring agent outputs are grounded in retrieved guideline context
- **Guideline Adherence:** Validation of recommendations against INASL standards

6.6.3 End-to-End Evaluation

- **Clinical Agreement Score:** Alignment with expert tumor board decisions
- **Completeness:** Coverage of relevant clinical factors in outputs
- **Clarity:** Readability and usability of generated reports

6.6.4 Safety Testing

- Evaluation of system ability to detect and flag clinically unsafe recommendations through human-in-the-loop validation

6.7 Future Evaluation Plan

A large-scale evaluation is planned as ongoing work, including:

- Expansion to 50–100 annotated patient cases
- Quantitative benchmarking across multiple clinical metrics
- Comparative analysis with tumor board decisions across institutions

6.8 Key Takeaway

Despite limited dataset size, the system demonstrates strong initial clinical validity, with high agreement to expert decisions and successful end-to-end execution in real-world scenarios. These results validate the feasibility of agent-based tumour board emulation and highlight its potential as a scalable and clinically grounded decision support tool.

Chapter 7

Contributions, Limitations and Future Work

7.1 Key Contributions

This project presents a comprehensive effort to design, implement, and deploy an AI-assisted tumour board emulation system. The key contributions of this work span technical, clinical, and system-level dimensions.

7.1.1 Technical Contributions

- **Modular Multi-Agent Architecture:** Designed and implemented a plug-and-play agent framework supporting modality-specific reasoning (clinical, radiology, pathology) and coordinated orchestration.
- **Knowledge Graph–Based Reasoning:** Introduced a hybrid reasoning layer that transforms structured patient data into a knowledge graph and applies ReAct-style planning with rule-based decision routing for context-aware agent invocation.
- **Schema-Driven AI Pipeline:** Developed a structured patient schema that acts as a unified interface across data ingestion, agent reasoning, and orchestration, enabling consistency and interpretability.
- **Guideline-Grounded Reasoning:** Integrated clinical guidelines (e.g., INASL) into agent outputs to ensure evidence-based recommendations and reduce hallucination risks.

7.1.2 System and Engineering Contributions

- **Full-Stack Clinical Platform:** Built a web-based dashboard (React + Python backend) supporting structured patient management, visualization, and AI interaction.
- **Secure Deployment and Access Control:** Implemented OAuth 2.0 authentication, JWT-based session management, role-based access control, and audit logging to support real-world clinical usage.
- **Efficient Agent Execution:** Designed a caching mechanism to store agent outputs, reducing redundant LLM calls and improving system efficiency.

- **PDF-to-Structured Data Pipeline:** Developed a data ingestion workflow that converts clinician-generated reports into structured schema representations, reducing manual effort.

7.1.3 Clinical and Workflow Contributions

- **Tumour Board Workflow Modeling:** Modeled real-world clinical decision-making processes and mapped them to agent orchestration and system design.
- **Explainable Decision Support:** Enabled structured and traceable outputs that align with tumour board discussions, improving interpretability and clinician trust.
- **Preliminary Clinical Validation:** Demonstrated approximately 90% agreement with clinician decisions on real-world cases, indicating strong alignment with expert reasoning.

7.2 Limitations and Challenges

Despite significant progress, several limitations remain that affect immediate scalability and deployment.

7.2.1 Limited Dataset and Evaluation Scale

The system was evaluated on a small number of de-identified cases due to constraints in data availability and clinician involvement. This limits statistical generalization and comprehensive benchmarking.

7.2.2 Dependence on Structured Input

The system's performance depends on the completeness and correctness of structured input data. Missing key parameters can impact the quality of reasoning despite system-level checks.

7.2.3 Non-Clinical Factors Not Modeled

Certain real-world considerations such as financial constraints, patient preferences, and institutional limitations are not explicitly represented in the current schema, leading to occasional mismatches with clinician decisions.

7.2.4 Language Model Limitations

LLM-based components remain susceptible to hallucination and variability in phrasing. While guideline grounding mitigates this risk, complete elimination of such issues remains an open challenge.

7.2.5 Clinical Validation Dependency

The system is designed as a decision support tool and requires clinician oversight for all recommendations. It is not intended to replace expert judgement.

7.3 Future Work

Building upon the current system, several directions are identified for further research and deployment.

7.3.1 Large-Scale Clinical Evaluation

Expanding the dataset to 50–100 annotated cases and conducting systematic benchmarking against tumour board decisions across institutions will enable robust validation.

7.3.2 Tumour Board Co-Pilot Deployment

Deploying the system alongside real tumour board meetings (e.g., PGIMER Chandigarh) in a co-pilot setting will allow evaluation of real-time utility, accuracy, and impact on clinician workflow.

7.3.3 Incorporation of Non-Clinical Factors

Future work will extend the schema and reasoning pipeline to incorporate patient-specific constraints such as financial considerations and treatment accessibility.

7.3.4 Model Optimization and Localization

Exploration of domain-specific and locally deployable models (e.g., Gemma, Sarvam AI) will improve efficiency, reduce dependency on external APIs, and enable deployment in resource-constrained environments.

7.3.5 Extension to Other Cancer Types

The modular schema and agent architecture can be extended to other tumour types such as colorectal, breast, and lung cancers with domain-specific adaptations.

7.4 Conclusion

This work demonstrates the feasibility of a secure, modular, and clinically grounded AI-assisted tumour board emulation system. By integrating multimodal data processing, knowledge-guided reasoning, and clinician-in-the-loop validation within a deployable framework, Samavet AI bridges the gap between experimental AI models and real-world clinical decision support.

The system not only replicates key aspects of multidisciplinary tumour board reasoning but also highlights the potential of agent-based AI systems to improve accessibility, consistency, and efficiency in oncology care. While further validation and scaling are required, the current work establishes a strong foundation for future research and clinical translation.

Chapter 8

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